

Causal Inference



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THE REMIX

Staggered DiD: Bacon + Callaway-Sant'Anna

Day 3 of 3: Morning

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Stevenson & Wolfers (2006), revisited

The question. Did no-fault unilateral divorce laws reduce female suicide rates? Treatment timing varies across states from 1969 to 1985.

The setup. Staggered adoption + two-way fixed effects:

$$y_{ist} = \alpha_i + \gamma_t + \delta D_{ist} + \varepsilon_{ist}$$

The headline result. Stevenson-Wolfers report a meaningful negative effect: unilateral divorce reduced female suicide.

Goodman-Bacon (2021) revisited it. That headline TWFE coefficient is **not** the ATT. It's a weighted average of every 2×2 DiD you could draw with the data — including 2×2 s where one already-treated state is used as the control for a later-treated state.

Goodman-Bacon, A. (2021), "Difference-in-differences with variation in treatment timing," J. Econometrics 225(2): 254–277.

A small simulation makes the problem visible

baker.do. 1,000 firms, 4 cohorts of states, treated at 1986/1992/1998/2004. *Dynamic* treatment effects: ATT grows each year since treatment.

	The truth	TWFE estimate
Mean ATT across cohorts and post-periods	+82	−6.69

The sign flipped.

- Same data. Same DGP. Same TWFE regression you've been running.
- No econometric error in the regression itself — OLS gave you exactly what OLS gives.
- Something is wrong with the *estimand* TWFE is targeting.

Simulation: the-remix-tour/simulations/staggered/baker.do; seed 20200403.

This morning, two acts

1. **Why TWFE goes wrong (Goodman-Bacon 2021).** TWFE is a variance-weighted average of every 2×2 DiD in the data. Some of those 2×2 s use *already-treated* units as controls. When effects are dynamic, that's poison.
2. **Callaway & Sant'Anna (2021), the fix.** Forward-engineer from the target parameter $ATT(g, t)$. Estimate each one with the never-treated or not-yet-treated — never the already-treated. Then aggregate.

The afternoon-morning thread: yesterday's DR-DiD is the engine inside CS's $ATT(g, t)$.

LESSON

1. Bacon's decomposition

TWFE is a weighted average of 2×2 DiDs — and OLS picks the weights

Bacon's claim: TWFE *is* a weighted average of 2×2s

The TWFE regression with K timing groups produces an estimate $\hat{\delta}^{\text{TWFE}}$ that is *numerically identical* to a weighted average of every 2×2 DiD you can construct from the data.

The decomposition (informally)

$$\hat{\delta}^{\text{TWFE}} = \sum_{\text{every } 2 \times 2 \text{ in the data}} s_{ij} \hat{\delta}_{ij}^{2 \times 2}$$

- This is *pure* OLS algebra. No assumptions about treatment effects yet.
- It's also not surprising: any OLS coefficient is a weighted average of some underlying “comparison” — Angrist (1998), Tymon (2022) for the general result.
- The interesting questions: **which 2×2s, with what weights, and is the average we get the average we want?**

With K timing groups, there are many 2×2 s

Three timing groups $\{a, b, c\}$ plus one never-treated group U . Bacon counts the 2×2 s:

Treated vs. Never	Treated vs. Later	Treated vs. Earlier
a vs. U	a vs. b (later)	b vs. a (earlier)
b vs. U	a vs. c (later)	c vs. a (earlier)
c vs. U	b vs. c (later)	c vs. b (earlier)

- Column 1: the comparisons you intended.
- Column 2: still uses an *eventually*-treated unit as a control — it's fine, because that unit hasn't been treated yet in the comparison window.
- Column 3: uses an *already*-treated unit as a control. **This is the forbidden comparison.**

Two timing groups: k (early) and ℓ (late)

Three time windows: PRE, MID (between the two treatment dates), POST.

- k vs. ℓ during MID. Group k is post-treatment; group ℓ is still pre-treatment. **Comparison is fine** — ℓ is a not-yet-treated control.

$$\widehat{\delta}_{k\ell}^{2 \times 2, k} = (\bar{y}_k^{\text{MID}} - \bar{y}_k^{\text{PRE}}) - (\bar{y}_\ell^{\text{MID}} - \bar{y}_\ell^{\text{PRE}})$$

- ℓ vs. k during POST. Group ℓ is post-treatment; group k is *also* post-treatment. **The forbidden comparison** — k is now an already-treated control.

$$\widehat{\delta}_{\ell k}^{2 \times 2, \ell} = (\bar{y}_\ell^{\text{POST}} - \bar{y}_\ell^{\text{MID}}) - (\bar{y}_k^{\text{POST}} - \bar{y}_k^{\text{MID}})$$

The full Bacon decomposition

The setup. K timing groups, possibly one never-treated group U . TWFE estimate:

$$\hat{\delta}^{\text{TWFE}} = \underbrace{\sum_{k \neq U} s_{kU} \hat{\delta}_{kU}^{2 \times 2}}_{\text{treated vs. never-treated}} + \underbrace{\sum_{k \neq U} \sum_{\ell > k} s_{k\ell} \left[\mu_{k\ell} \hat{\delta}_{k\ell}^{2 \times 2, k} + (1 - \mu_{k\ell}) \hat{\delta}_{\ell k}^{2 \times 2, \ell} \right]}_{\text{treated vs. another-timing-group}}$$

- First sum: every timing group compared to the never-treated. Clean.
- Second sum: every *pair* of timing groups (k, ℓ) with k earlier, ℓ later, contributes two 2×2 s — one with ℓ as not-yet-treated control (fine), one with k as already-treated control (**forbidden**).
- Weights s_{kU} , $s_{k\ell}$, and the within-pair weight $\mu_{k\ell}$ come from OLS algebra, not theory.

Where the weights come from: variance of $\tilde{D}$

After de-meaning D_{it} within unit and time, what remains is $\tilde{D}_{it}$. The Bacon weights are functions of $\widehat{\text{Var}}(\tilde{D}_{it})$ and group shares:

$$s_{kU} = \frac{n_k n_U \bar{D}_k (1 - \bar{D}_k)}{\widehat{\text{Var}}(\tilde{D}_{it})}$$

$$s_{k\ell} = \frac{n_k n_\ell (\bar{D}_k - \bar{D}_\ell) (1 - (\bar{D}_k - \bar{D}_\ell))}{\widehat{\text{Var}}(\tilde{D}_{it})}$$

$$\mu_{k\ell} = \frac{1 - \bar{D}_k}{1 - (\bar{D}_k - \bar{D}_\ell)}$$

- $\bar{D}_k$ is the share of group k 's panel spent in treatment.
- Same variance-weighting Angrist (1998) and Tymon (2022) describe generally: OLS weights treatment contrasts by treatment-status variance.

Variance is maximized in the middle of the panel

Consider the never-treated 2×2 weight, s_{kU} . The key factor: $\overline{D}_k(1 - \overline{D}_k)$.

$\overline{D}_k$ (share of panel treated)	$\overline{D}_k(1 - \overline{D}_k)$
0.1	0.09
0.3	0.21
0.5	0.25 ← max
0.7	0.21
0.9	0.09

- Groups treated in the *middle* of the panel get the highest weight.
- Same logic for $(\overline{D}_k - \overline{D}_\ell)(1 - (\overline{D}_k - \overline{D}_\ell))$.
- **Panel length affects the weights.** Trim the panel and you've changed the estimand.

So far: pure algebra. Now: potential outcomes.

What we have:

- TWFE is a weighted average of 2×2 s.
- Some of those 2×2 s use already-treated units as controls.
- OLS picks the weights based on treatment variance.

What we don't yet have:

- Whether the weighted average we get *is* the weighted ATT we want.

To answer that we need to switch from realized outcomes to potential outcomes.

The next four slides walk through a single 2×2 step by step, then substitute back into the decomposition. The result is a clean expression for the bias.

LESSON

2. The forbidden 2×2 , step by step

Add a zero, rearrange, and the bias appears

Step 1 — write down the 2×2 in observed outcomes

The 2×2 we're worried about: group ℓ (later-treated) in POST, with group k (earlier-treated) as control.

$$\hat{\delta}_{\ell k}^{2 \times 2, \ell} = [\bar{y}_{\ell}^{\text{POST}} - \bar{y}_{\ell}^{\text{MID}}] - [\bar{y}_k^{\text{POST}} - \bar{y}_k^{\text{MID}}]$$

- Both groups are *treated* in POST.
- Both groups were *treated* in MID? No — ℓ is still untreated in MID.
- So group ℓ 's “change” includes its move from untreated to treated. Group k 's “change” is from *treated* to *treated*.

Step 2 — switch to potential outcomes

Replace each observed mean with the appropriate potential-outcome mean (using the switching equation):

$$\bar{y}_{\ell}^{\text{POST}} = \mathbb{E}[Y^1 \mid \ell, \text{POST}] \quad (\text{treated})$$

$$\bar{y}_{\ell}^{\text{MID}} = \mathbb{E}[Y^0 \mid \ell, \text{MID}] \quad (\text{not yet treated})$$

$$\bar{y}_k^{\text{POST}} = \mathbb{E}[Y^1 \mid k, \text{POST}] \quad (\text{treated})$$

$$\bar{y}_k^{\text{MID}} = \mathbb{E}[Y^1 \mid k, \text{MID}] \quad (\text{already treated})$$

Notice the asymmetry. Group k is in Y^1 in *both* MID and POST. That's the problem.

Three of the four cells are Y^1 . Only one is Y^0 .

Step 3 — add zeros

We add three zero-valued terms of the form $\mathbb{E}[Y^0 \mid \cdot] - \mathbb{E}[Y^0 \mid \cdot]$:

The three zeros

1. $+\mathbb{E}[Y^0 \mid \ell, \text{POST}] - \mathbb{E}[Y^0 \mid \ell, \text{POST}]$ (add to ℓ in POST)
2. $+\mathbb{E}[Y^0 \mid k, \text{POST}] - \mathbb{E}[Y^0 \mid k, \text{POST}]$ (add to k in POST)
3. $+\mathbb{E}[Y^0 \mid k, \text{MID}] - \mathbb{E}[Y^0 \mid k, \text{MID}]$ (add to k in MID)

- Each adds zero. The 2×2 is unchanged.
- But now we can group the terms into something interpretable.

Step 4 — rearrange into ATT, parallel-trends, and ΔATT

Group the eight terms (four observed + four added zeros) into three buckets:

The forbidden 2×2 in causal pieces

$$\underbrace{\text{ATT}_\ell(\text{POST})}_{\text{what we want}} + \underbrace{\Delta Y_\ell^0(\text{POST}, \text{MID}) - \Delta Y_k^0(\text{POST}, \text{MID})}_{\text{parallel-trends bias}} - \underbrace{[\text{ATT}_k(\text{POST}) - \text{ATT}_k(\text{MID})]}_{\text{heterogeneity bias: } \Delta\text{ATT}_k} = \widehat{\delta}_{\ell k}^{2 \times 2, \ell}$$

- Bucket 1 is the ATT we want for group ℓ in the post-period.
- Bucket 2 is the standard PT bias — requires PT between ℓ and k in this window.
- Bucket 3 is **new**: it's the *change* in group k 's own ATT between MID and POST. If treatment effects grow (dynamics), this is positive — and it enters with a *minus sign*.

Step 5 — substitute back into Bacon's decomposition

The three flavors of 2×2 in causal pieces:

$$\begin{aligned}\hat{\delta}_{kU}^{2 \times 2} &= \text{ATT}_k(\text{POST}) + \Delta Y_k^0(\text{post}, \text{pre}) - \Delta Y_U^0(\text{post}, \text{pre}) \\ \hat{\delta}_{k\ell}^{2 \times 2, k} &= \text{ATT}_k(\text{MID}) + \Delta Y_k^0(\text{MID}, \text{pre}) - \Delta Y_\ell^0(\text{MID}, \text{pre}) \\ \hat{\delta}_{\ell k}^{2 \times 2, \ell} &= \text{ATT}_\ell(\text{POST}) + \Delta Y_\ell^0(\text{POST}, \text{MID}) - \Delta Y_k^0(\text{POST}, \text{MID}) \\ &\quad - \Delta \text{ATT}_k\end{aligned}$$

Plug these into Bacon's weighted sum. The TWFE probability limit is:

$$\text{plim } \hat{\delta}^{\text{TWFE}} = \text{VWATT} + \text{VWPT} - \Delta \text{ATT}$$

where VW means “variance-weighted” using the Bacon weights.

What the decomposition tells us

$$\text{plim } \hat{\delta}^{\text{TWFE}} = \text{VWATT} + \text{VWPT} - \Delta\text{ATT}$$

- **VWATT** — a variance-weighted ATT. Not the ATT *you* want unless you specifically wanted variance-weighted averaging. The weights depend on *panel length* (a researcher choice).
- **VWPT** — variance-weighted parallel-trends bias. Zero only if PT holds across *every* pairwise comparison, including between treated groups.
- **ΔATT** — the dynamics term. If ATT grows with time-since-treatment, this is *positive* and enters with a *minus sign* — attenuating TWFE toward zero, or even **flipping the sign**.

That's exactly what happens in baker.do: dynamics big enough that $-\Delta\text{ATT}$ swamps the rest, flipping +82 to -6.69.

LESSON

3. baker.do, in code

Set the seed, write the DGP, compute the truth, run TWFE, watch it fail

The DGP in 12 lines (baker.do)

```
* 40 states, 25 firms per state, 30 years (1980-2009)
* 4 cohorts treated at 1986, 1992, 1998, 2004
set seed 20200403
* ... set up the panel (state, firms, year) ...

* Group-specific TE means (heterogeneous across cohorts)
gen te1 = rnormal(10, 0.04) // cohort 1986
gen te2 = rnormal( 8, 0.04) // cohort 1992
gen te3 = rnormal( 6, 0.04) // cohort 1998
gen te4 = rnormal( 4, 0.04) // cohort 2004
gen te = cond(group==1, te1, cond(group==2, te2, cond(group==3, te3, te4)))

* Y(0) and Y(1) under DYNAMIC effects (te grows since treatment)
gen y0 = firms + n + e
gen te_dynamic = te*treat*(year - treat_date + 1)
gen y = y0 + te_dynamic
```

One step per line, named variables, no loops. simulations/staggered/baker.do.

The truth, computed before running anything

```
* The truth (averaged across cohorts and post-periods)  
su te_dynamic if treat == 1  
display "True average ATT (dynamic) = " r(mean)
```

Output: True average ATT (dynamic) = 82

Why 82? The dynamic effect for group 1 averages $(10 + 20 + \dots + 240)/24$. Combined across cohorts and post-periods, the simple average is 82.

We computed the truth from the DGP. Now compare it to what TWFE produces.

TWFE, applied

```
areg y i.year treat, a(id) robust
```

	Truth	TWFE
$\widehat{ATT}$	+82	−6.69***

Sign-flipped. Extreme dynamics — the $-\Delta ATT$ term swamps VWATT.

The Bacon decomposition of that -6.69

DD comparison	Weight	Avg DD estimate
Earlier-treated vs. later-treated-(as-control)	0.500	+51.80
Later-treated vs. earlier-treated-(as-control)	0.500	-65.18

$$(0.5)(+51.80) + (0.5)(-65.18) = -6.69.$$

Read it. Half the weight is on a 2×2 averaging -65 — a 2×2 that uses an already-treated group as control. **That's the forbidden comparison doing the damage.**

LESSON

4. Visualizing staggered structure

What the cells look like, what TWFE averages over

Constant-TE worksheet

Year	ATT(1986,t)	ATT(1992,t)	ATT(1998,t)	ATT(2004,t)
1980–1985	0	0	0	0
1986–1991	10	0	0	0
1992–1997	10	8	0	0
1998–2003	10	8	6	0
2004–2009	10	8	6	4
ATT(<i>g</i>)	10	8	6	4

- **Simple ATT** (uniform avg over colored cells) = 8.0.
- **VWATT** (TWFE's target) $\neq$ 8.0 — OLS weights down edge cohorts via $\bar{D}(1 - \bar{D})$.

Even with constant effects per cohort, TWFE doesn't give the simple ATT.

Dynamic-TE worksheet

Year	ATT(1986,t)	ATT(1992,t)	ATT(1998,t)	ATT(2004,t)
1986	10	0	0	0
1992	70	8	0	0
1998	130	56	6	0
2004	190	104	42	4
2009	240	144	72	24

ATT grows each year since treatment (the dynamic DGP). Simple ATT (uniform avg over colored cells) ≈ 82 ; TWFE = -6.69 .

The $-\Delta$ ATT bias is green doing the damage. Late cohorts use the already-treated earlier cohorts as “controls” — whose outcomes are growing.

LESSON

5. Callaway & Sant'Anna

Forward-engineer from $ATT(g, t)$, never use the already-treated as control

The forward-engineering principle

Bacon told us what reverse-engineering from TWFE buys you: a variance-weighted average of every 2×2 in the data, including the forbidden ones.

The Callaway-Sant'Anna (2021) inversion:

1. **Start** with the target parameter you actually want: $ATT(g, t)$ — the average treatment effect on cohort g in calendar period t .
2. **State** the identification assumptions formally.
3. **Derive** an estimator directly from those assumptions.
4. **Aggregate** the building blocks into whatever summary you want.

Each $ATT(g, t)$ is its own clean 2×2 problem. Yesterday afternoon's DR-DiD is exactly the engine that estimates one.

ATT(g, t) — the building block

Definition

$$\text{ATT}(g, t) = \mathbb{E}[Y_{i,t}(1) - Y_{i,t}(0) \mid G_i = g]$$

- $G_i = g$ = “unit i was first treated in period g .” (Cohort identifier.)
- One $\text{ATT}(g, t)$ for every (cohort, calendar-period) pair where the cohort is treated.
- In baker.do: 4 cohorts $\times$ (varying numbers of post-periods) = 60 $\text{ATT}(g, t)$ cells in the grid.
- Each cell is a single 2×2 estimand. **The aggregation is a separate, downstream choice.**

Three flavors of parallel trends for staggered settings

The 2×2 PT assumption generalizes three ways:

Assumption	Comparison group	When to use
PT-GT-Nev	Never-treated only	Conservative; needs a never-treated to exist
PT-GT-NYT	Not-yet-treated	Authors' preferred — middle ground
PT-GT-all	All groups, all periods	Strongest assumption; imposes parallel pre-trends

- NYT lets late-treated groups serve as controls for earlier-treated groups in their pre-treatment window. Never as already-treated controls.
- The forbidden 2×2 of Bacon is *ruled out by construction*.

Baker, Callaway, Cunningham, Goodman-Bacon, Sant'Anna (2025), "DiD: A Practitioner's Guide."

The estimator — and yesterday's DR is the recommended one

For each $ATT(g, t)$, three options for the building-block estimator:

1. **Outcome Regression (OR).** Heckman-Ichimura-Todd: model $\Delta Y(0)$ on X using controls, impute the treated counterfactual.
2. **Inverse Propensity Weighting (IPW).** Abadie (2005): weight controls by $p(X)/(1 - p(X))$.
3. **Doubly Robust (DR).** Sant'Anna-Zhao (2020). Consistent if *either* the OR model or the pscore model is correct.

The practitioner's-guide recommendation

“Favor doubly robust over RA or IPW alone.”

The DR-DiD we proved yesterday is exactly the engine sitting inside each cell of CS.

Aggregation is a downstream choice, not an estimation choice

Once you have $\{\widehat{ATT}(g, t)\}$, build whatever summary the policy question demands:

- **Simple ATT.** Uniform average over all post-treatment (g, t) cells.
- **Group ATT, $ATT(g)$.** Average each cohort's own post-treatment cells. "What did the policy do to cohort g on average?"
- **Event-time ATT, $ATT_{es}(e)$.** Average across cohorts at fixed event time e since treatment. "What's the dynamic profile?"
- **Calendar-time ATT.** Average across cohorts within a calendar period. "What happened in 2002, on average?"

Inference. Bootstrap, with uniform confidence bands when reporting an event-study (the `did` R package does this by default).

baker_cs.do — the implementation

```
* Get the same baker data, fit CS with DR-DiD
ssc install csdid, replace
ssc install drdid, replace

csdid y, ivar(id) time(year) gvar(treat_date) ipw long2

* Four aggregations, four answers:
csdid_estat simple // simple ATT across all cells
csdid_estat group // ATT(g) per cohort
csdid_estat calendar // ATT by calendar period
csdid_estat event // event-time profile
```

Building blocks first, aggregation second.

collegio-carlo-alberto/labs/Baker/Solutions/baker_cs.do.

The headline: CS recovers the truth

	Truth	TWFE	CS (DR-DiD)
Feasible ATT (pre-2004 sample)	68.33	26.81***	68.34***

- The “feasible” sample drops the 2004 cohort so a never-treated comparison exists.
- TWFE is no longer sign-flipped (no universally-treated tail), but it still attenuates — still has VWATT weights, still uses early-treated cohorts as controls.
- CS lands on the truth to two decimals.

Not every $ATT(g, t)$ is feasible

CS can only estimate $ATT(g, t)$ if there is a valid comparison group for (g, t) :

- Under **PT-GT-Nev**: need a never-treated unit observed at t .
- Under **PT-GT-NYT**: need a not-yet-treated unit observed at t .

In baker.do: The last cohort is treated in 2004. After 2004, every unit is treated. **No comparison units exist for $ATT(g, t > 2004)$.**

Two responses:

- Restrict to the feasible sample. CS will tell you which (g, t) it dropped.
- Or, if you really want estimates for the universally-treated tail, use a different framework (Imputation: Borusyak et al. 2024; Wooldridge 2021) that doesn't require a comparison group.

LESSON

6. Synthesis

What's the same, what's different, what to do tomorrow morning

Yesterday afternoon plus this morning = one workflow

- **Yesterday (covariates).** Conditional PT $\Rightarrow$ DR-DiD. Two chances to be right (OR model or pscore model).
- **This morning (staggered).** $ATT(g, t)$ building blocks, estimated cell-by-cell, then aggregated.
- **The combination.** Each $ATT(g, t)$ is a 2×2 problem. The recommended way to estimate it is DR-DiD with whatever X you need for conditional PT. Then aggregate.

One sentence

Forward-engineer the target, estimate the cells with DR, aggregate with non-negative weights.

Basically never.

- TWFE = ATT only when (a) treatment effects are homogeneous across cohorts, AND (b) treatment effects don't change over time within a cohort.
- Both fail in almost every applied setting.
- Even when they hold, the variance weights are still strange (depend on panel length).
- **If you must report it, also report CS, Sun-Abraham, or imputation, and explain the differences.**

Baker, Callaway, Cunningham, Goodman-Bacon, Sant'Anna (2025): "Do not use TWFE."

A student asks: “But what about my paper from last year that used TWFE?”

A reasonable thing to do, in order:

1. **Run the Bacon decomposition** (`ddtiming` in Stata, `bacondecomp` in R). See whether the forbidden 2×2 s carry meaningful weight.
2. **Re-estimate with CS or Sun-Abraham.** Compare the headline to TWFE.
3. **If they disagree:** the difference *is* the $-\Delta\text{ATT}$ or VWPT contamination. Trust the design-based estimator.
4. **If they agree:** in your specific setting, the dynamics and heterogeneity didn't bite. Report both, explain.

Goodman-Bacon, Sant'Anna, Whitten (2026) gives a more refined re-analysis framework for pre-trends-parallel settings.

Lab: rerun baker, rerun baker_cs, compare

- **Step 1.** Run `simulations/staggered/baker.do` to (re)generate the dynamic-TE dataset. Confirm the truth is +82 and TWFE is -6.69 .
- **Step 2.** Run the Bacon decomposition (`ddtiming`). Verify the two weights and the two 2×2 averages reproduce the textbook table.
- **Step 3.** Run `baker_cs.do`: CS with DR-DiD, plus four aggregations (`simple`, `group`, `calendar`, `event`).
- **Step 4.** Drop the 2004 cohort. Re-run TWFE: notice it stops sign-flipping (no longer have an universally-treated tail). Re-run CS: now lands on the truth.
- **Step 5.** Write a one-paragraph summary: which assumption is doing the work, and what would have to be true for TWFE to be correct?

Walkthrough: `zurich/labs/Baker/README.md`. Code: `simulations/staggered/baker.do`, plus the CS files Scott will provide.

Tomorrow

- Sensitivity analysis: how robust is CS to PT failing? (Rambachan-Roth.)
- Honest event-studies and how to read pre-trends without fooling yourself.
- Continuous and multi-valued treatment: when “dose” matters.

Reading for tonight: *Baker, Callaway, Cunningham, Goodman-Bacon, Sant’Anna (2025)*
“Difference-in-Differences Designs: A Practitioner’s Guide.”

Forward-engineer.

*Start with what you want to estimate.
Match the estimator to the assumption.
Aggregate at the end, not the start.*